Population-scale coupling between electrotonic regime and synaptic spatial organization in mouse visual cortex

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Abstract

Dendritic branches differ in their capacity to electrically isolate synaptic inputs, ranging from electrotonically compact domains where inputs sum linearly to compartmentalized domains capable of local nonlinear integration. Whether this biophysical heterogeneity is reflected in the spatial arrangement of synapses remains poorly understood, in part because prior evidence has been limited to small samples and shallow mechanistic tests. Here we report a population-scale analysis of 2,138 proofread neurons (136,414 dendritic branches, ∼8.2 million synapses) from the MICrONS mouse visual cortex electron microscopy volume. Using continuous electrotonic length (L/λ) as a predictor in mixed-effects models, we find that synaptic spatial organization covaries with dendritic biophysical regime: electrotonically isolated branches carry more structured synaptic arrangements, as measured by two independent spacing metrics (Clark-Evans z: inhibitory $\beta = 7.65$, $p < 10^{-77}$; Interval CV z: inhibitory $\beta = -17.5$, $p < 10^{-90}$). This coupling is robust to threshold choice (8/8 discretization schemes converge) and is not explained by excitatory/inhibitory input balance (mediation < 7%). Partner architecture features — partner diversity and contact multiplicity — mediate 25–53% of the coupling at population scale, identifying two distinct wiring mechanisms: partner diversity governs local spacing regularity while contact multiplicity governs global synapse dispersion. Random-slope mixed models reveal that excitatory and inhibitory neurons implement fundamentally different organizational logics: excitatory neurons show steep $L/\lambda \rightarrow$ spacing regularity slopes (median $= -2.23$) while inhibitory neurons maintain near-uniform spacing regardless of electrotonic regime (median $= -0.07$; Cohen's $d = 0.83$, $p = 1.1 \times 10^{-24}$). We decompose the inhibitory residual — the ∼86% of inhibitory coupling not explained by partner architecture — by tracing afferent cell type identity through a multi-stage pathway. Basket cell (BC) afferents are strongly enriched on compact branches (neuron-level median $\log_2$ enrichment $= 0.99$, 89% of neurons positive, $p = 4.0 \times 10^{-45}$), concentrated in proximal dendritic compartments. Afferent identity and partner architecture jointly explain 18–26% of the inhibitory regime effect, reducing the residual from ∼86% to ∼75%. Critically, within-axon bouton placement analysis reveals that BC contacts are placed more compactly on compact branches even after conditioning on contact multiplicity (regime coefficient $= +0.95$, $p = 1.1 \times 10^{-107}$; equal-k subsets: $p < 10^{-18}$ at every k), and that Martinotti cell (MC) contacts spread more broadly than BC contacts on compartmentalized branches (interaction $p = 5.2 \times 10^{-4}$). These results establish a complete mechanistic chain — electrotonic regime → afferent subtype composition → partner architecture → within-axon placement → synaptic spacing — and identify subtype-specific placement rules as a previously uncharacterized layer of inhibitory circuit organization.

1 Introduction

Dendritic trees are not passive conduits. The electrotonic properties of a branch — its length, diameter, and distance from the soma — determine how synaptic inputs are integrated, from linear summation in compact domains to supralinear amplification in electrically isolated compartments (Rall, 1967; London and Häusser, 2005; Koch, 1999). This biophysical heterogeneity has long

been recognized as the basis for dendritic computation: thin, isolated branches can function as independent computational subunits, generating local NMDA spikes that are thresholded before reaching the soma (Mel, 1994; Polsky et al., 2004; Poirazi et al., 2003; Larkum et al., 2009).

Most connectomics studies characterize synaptic connectivity in terms of *who* connects to *whom* — connection probabilities, motif statistics, cell-type circuit diagrams. A complementary question, less frequently addressed, is what generative rules produce the *spatial* architecture of connections on dendrites. Spatial statistics on synapse positions can, in principle, reveal these rules: if the arrangement of synapses varies systematically with the biophysical properties of the branch, and if that variation can be decomposed into contributions from specific wiring features, then the spatial pattern encodes information about the mechanisms that generated it.

If spatial organization of synaptic inputs encodes something about dendritic computation, the arrangement of synapses should vary systematically with the electrotonic regime of the branch they innervate. Clustered inputs on isolated branches would support nonlinear integration; distributed inputs on compact branches would be consistent with linear summation (Branco et al., 2010; Mel, 1994). Alternatively, if spatial patterns reflect wiring logistics — which axons can physically reach which dendritic domains, and how many contacts they make — the coupling between regime and spatial organization would be explained by partner-level innervation structure rather than computational optimization (Voruganti, 2026).

A prior study on 12 manually curated MICrONS neurons reported preliminary evidence for the latter hypothesis: partner contact multiplicity mediated 69% of the coupling between electrotonic regime and synaptic compactness in excitatory neurons, with only 39% mediation in inhibitory neurons (Voruganti, 2026). However, the small sample limited statistical power and the precision of these estimates: one of three spatial metrics failed to reach significance, per-neuron consistency was low (2/12 individually significant), and the inhibitory subtype decomposition rested on $n = 2$ per subtype. Three critical questions remained unanswered: does partner-level mediation replicate at population scale? Does the $L/\lambda \rightarrow$ spatial coupling slope vary across neurons, or is it a fixed population property? And what explains the non-monotonic dose-response in pairwise compactness?

Here we address these questions by scaling the analysis to all 2,138 proofread neurons in the MICrONS volume (136,414 branches, ∼8.2 million synapses). We compute per-branch partner architecture features across the full population, fit random-slope mixed models that allow the coupling to vary per neuron, and resolve the compactness non-monotonicity through geometric diagnostics. We find that partner architecture mediates 25–53% of the coupling (exceeding the shallow 1–10% mediation by input balance), that excitatory and inhibitory neurons implement qualitatively different organizational logics, and that the compactness non-monotonicity is a normalization artifact rather than a biological transition zone.

2 Results

2.1 Population-scale morphometric analysis

We extracted dendritic skeletons, synapse coordinates, and presynaptic partner identities for all 2,138 proofread neurons in the MICrONS mouse visual cortex volume (MICrONS Consortium, 2021; Turner et al., 2022) using the CAVEclient programmatic interface. Neurons spanned 11 cell types: seven excitatory subtypes (23P, $n = 518$; 4P, $n = 508$; 5P-ET, $n = 70$; 5P-IT, $n = 195$; 5P-NP, $n = 13$; 6P-CT, $n = 164$; 6P-IT, $n = 234$) and four inhibitory subtypes (basket cells [BC], $n = 213$; Martinotti cells [MC], $n = 114$; bipolar cells [BPC], $n = 85$; neurogliaform cells [NGC],

$n = 24$). We identified 136,414 dendritic branches carrying a total of approximately 8.2 million synapses.

For each branch, we computed electrotonic length L/λ using cable theory with standard passive parameters ($R_M = 20{,}000\ \Omega \cdot \text{cm}^2$, $R_I = 150\ \Omega \cdot \text{cm}$) and branch-specific diameters derived from per-node radii in the EM reconstruction (Methods). The electrotonic length distribution was right-skewed (median $L/\lambda = 0.059$, IQR = 0.024–0.121), with 62.4% of branches below $L/\lambda = 0.1$ (electrotonically compact) and 4.5% above $L/\lambda = 0.5$ (strongly compartmentalized). For discretized analyses, we classified branches into three regimes using tertile thresholds on the L/λ distribution (regime 0: $L/\lambda < 0.037$; regime 1: $0.037 \leq L/\lambda < 0.157$; regime 2: $L/\lambda \geq 0.157$).

We quantified synaptic spatial organization using three complementary metrics, each z-scored against 1,000 uniform null draws per branch (Methods). The first two assess *local spacing structure* — whether neighboring synapses are evenly distributed along the branch: the **Clark-Evans ratio** (Clark and Evans, 1954) compares the mean nearest-neighbor distance to that expected under uniform placement, and the **Interval CV** measures the regularity of the gaps between consecutive synapses. The third assesses *global dispersion* — how spread out the entire synapse population is across the branch: **pairwise compactness** is the mean Euclidean distance between all synapse pairs. These two dimensions — local spacing and global dispersion — need not covary, a distinction that becomes important in Sections 2.5 and 2.7. Positive Clark-Evans z indicates more regular spacing than expected by chance; negative Interval CV z indicates more uniform inter-synapse intervals; positive compactness z indicates tighter clustering (Figure 1).

2.2 Continuous electrotonic length is associated with synaptic spatial organization

To avoid discretization artifacts, we first tested whether spatial organization covaries continuously with electrotonic length. We fit linear mixed-effects models with L/λ as a continuous predictor, controlling for synapse count and total branch length, with neuron identity as a random intercept (Methods).

Inhibitory neurons showed strong continuous coupling across all three metrics: Clark-Evans z increased with L/λ ($\beta = 7.65$, $p = 2.3 \times 10^{-77}$, $n = 22{,}808$ branches), Interval CV z decreased ($\beta = -17.5$, $p = 1.4 \times 10^{-90}$), and pairwise compactness z increased ($\beta = 11.7$, $p = 9.1 \times 10^{-117}$). The direction is consistent: longer, more isolated branches carry synapses that are more regularly spaced, with more uniform intervals, and that are on average more tightly clustered (Figure 2).

Excitatory neurons showed a more nuanced pattern. Interval CV z scaled strongly with L/λ ($\beta = -3.10$, $p = 1.0 \times 10^{-67}$, $n = 89{,}195$ branches), and Clark-Evans z showed a weaker but significant relationship ($\beta = -0.63$, $p = 2.5 \times 10^{-5}$, $n = 86{,}595$ branches). The negative sign of the excitatory Clark-Evans coefficient, opposite to the positive inhibitory coefficient, does not reflect a reversal of the biological direction: both cell types show more regular spacing on isolated branches. The sign difference arises because the mixed-effects model adjusts for covariates (synapse count, branch length) that shift the baseline z-score distributions differently in excitatory and inhibitory populations (Figure 7). However, pairwise compactness showed no significant continuous relationship ($\beta = -0.07$, $p = 0.67$, $n = 89{,}200$ branches). This metric-specific divergence is resolved in Section 2.6, where random-slope models reveal that it reflects fundamentally different organizational logics rather than a simple absence of coupling.

2.3 Discretized regime analysis confirms graded coupling

To contextualize the continuous results and enable comparison with prior work, we classified branches into three electrotonic regimes using tertile thresholds and fit mixed-effects models with regime as a categorical predictor, controlling for synapse count, total branch length, and excitatory fraction of presynaptic inputs.

All three metrics showed large regime effects (Figure 3):

- Clark-Evans z: regime 2 coefficient = 0.74 (Cohen's $d = 0.60$; $n = 109{,}403$ branches)
- Interval CV z: regime 2 coefficient = -1.70 (Cohen's $d = -0.64$; $n = 112{,}202$)
- Pairwise compactness z: regime 2 coefficient = 0.91 (Cohen's $d = -0.07$; $n = 112{,}207$)

All regime effects were significant at $p < 10^{-300}$.

Dose-response analysis confirmed that the Clark-Evans and Interval CV effects are monotonically graded across regimes (Spearman $\rho = 0.25$ and $\rho = -0.37$, respectively; $p < 10^{-200}$ for both). The median Clark-Evans z progresses from -0.15 (regime 0) to 0.37 (regime 1) to 0.61 (regime 2), and the median Interval CV z from -0.05 to -0.78 to -1.14.

Sensitivity to threshold choice. We tested the robustness of the discretized result across eight threshold schemes: the primary tertile thresholds, biophysical thresholds ($L/\lambda = 0.1, 0.5$), and six quantile-based variants spanning 20th/80th to 40th/60th percentile splits. All eight converged for all three metrics (24/24 models), with Clark-Evans regime 2 coefficients ranging from 0.55 (biophysical) to 0.80 (20th/80th quantile) (Figure 1B). This stability demonstrates that the coupling is not an artifact of a particular threshold choice.

2.4 Cell-type dissociation: inhibitory coupling is uniformly stronger

The cell-type dissociation identified in prior work (Voruganti, 2026) replicates at population scale and extends to all four inhibitory subtypes tested (Figure 4).

For Clark-Evans z, inhibitory neurons show a regime 2 coefficient of 1.32 (Cohen's $d = 0.94$, $n = 22{,}808$) compared to 0.69 for excitatory neurons ($d = 0.60$, $n = 86{,}595$) — a 1.9-fold difference. This pattern holds uniformly across inhibitory subtypes:

- **Basket cells** ($n = 213$, 13,944 branches): regime 2 coefficient = 1.68, $d = 1.15$
- **Bipolar cells** ($n = 85$, 3,573 branches): regime 2 coefficient = 1.16, $d = 0.83$
- **Neurogliaform cells** ($n = 24$, 1,693 branches): regime 2 coefficient = 1.08, $d = 0.39$
- **Martinotti cells** ($n = 114$, 7,438 branches): regime 2 coefficient = 0.79, $d = 0.64$

Basket cells show the strongest coupling of any cell type, with effect sizes exceeding all excitatory subtypes (excitatory range: $d = 0.45$–1.14 across seven subtypes). Martinotti cells, despite being the weakest inhibitory subtype, still show coupling comparable to the excitatory average. Within excitatory neurons, layer 5 extratelencephalic-projecting neurons (5P-ET, $d = 1.00$) and layer 6 corticothalamic neurons (6P-CT, $d = 0.91$) show particularly strong effects, while layer 2/3 pyramidal cells (23P, $d = 0.50$) and layer 4 pyramidal cells (4P, $d = 0.45$) are weaker.

The strength hierarchy across subtypes — BC > BPC > NGC > MC among inhibitory, 5P-ET > 6P-CT > 5P-IT > 6P-IT > 23P > 4P among excitatory — is consistent across metrics and suggests that the magnitude of regime coupling reflects cell-type-specific wiring architecture. The biological basis of this hierarchy is addressed in Section 2.6.

2.5 Partner architecture mediates regime coupling via two distinct mechanisms

The original 12-neuron study identified partner contact multiplicity as the primary mediator of regime coupling (69% mediation in excitatory neurons; Voruganti 2026). However, the population-scale replication (Section 2.8) found that excitatory/inhibitory input balance — the only partner-level feature tested at scale — mediates only 1–10% of the coupling, leaving a large mechanistic gap. We addressed this by computing four partner architecture features for each of 127,697 branches across all 2,138 neurons (Methods).

Per-branch partner architecture features. For each branch with matched presynaptic partners, we computed: (1) **unique partners**, the number of distinct presynaptic neurons; (2) **mean synapses per partner**, the ratio of total synapse count to unique partners (contact multiplicity); (3) **Gini coefficient of partner synapse counts**, measuring concentration inequality across partners; and (4) **fraction of multi-synaptic partners**, the proportion of partners making ≥ 2 contacts on the branch. Verification confirmed perfect correlation between the product of unique partners and mean synapses per partner and total synapse count ($r = 1.000$), validating the feature computation.

Individual mediator analysis. We tested each partner feature as a mediator using the coefficient-comparison framework (Methods). The results revealed a striking dissociation between which features mediate which spatial dimensions (Table 2):

Partner diversity governs local spacing. For Clark-Evans z and Interval CV z — the two metrics that assess nearest-neighbor regularity and interval uniformity — **unique partners** was the strongest individual mediator in the overall population (23.0% and 22.9% mediation, respectively). Within excitatory neurons specifically, fraction of multi-synaptic partners was marginally stronger for Clark-Evans (23.7% vs. 22.7%), but the difference is within noise and both features reflect the same underlying partner diversity mechanism. This indicates that the number of distinct presynaptic neurons innervating a branch is the primary determinant of how evenly synapses are spaced.

Contact multiplicity governs global dispersion. For pairwise compactness z — the metric that assesses how spread out synapse are across the branch — **mean synapses per partner** was the strongest individual mediator (39.3% overall). This indicates that how many contacts each partner makes determines the spatial concentration of the overall synapse population.

All four partner features were individually significant mediators (all $p < 10^{-77}$), confirming that partner architecture operates through multiple correlated but non-redundant channels.

Combined mediation. When all four partner features were included simultaneously, combined mediation reached 28.9% for Clark-Evans z, 25.2% for Interval CV z, and 43.3% for pairwise compactness z in the overall population. This far exceeds the 1–10% mediation by input balance alone, establishing partner architecture as the dominant mediator of regime coupling at population scale.

Cell-type stratification reveals the inhibitory residual. The most important finding is the asymmetry between cell types. In excitatory neurons, partner architecture mediates 29–53% of the coupling (combined: 29.4% Clark-Evans, 31.3% Interval CV, 52.6% compactness). This is consistent with the original 12-neuron estimate of 69% mediation in excitatory neurons, converging from a larger population with more features.

In inhibitory neurons, by contrast, partner architecture mediates only 12–14% (combined: 14.0% Clark-Evans, 12.1% Interval CV, 12.6% compactness). This means that while inhibitory neurons show ∼2× stronger regime coupling than excitatory neurons (Section 2.4), substantially less of that coupling is explained by the wiring features we can measure. Something else — beyond partner diversity and contact multiplicity — is organizing inhibitory dendrites. We term this the **inhibitory residual**: the ∼86% of inhibitory coupling that cannot be attributed to partner-level innervation structure. Its identity is an open question addressed in the Discussion.

2.6 Excitatory and inhibitory neurons implement distinct organizational logics

The analyses above treat the $L/\lambda \rightarrow$ spatial coupling as a fixed population-level property, with neuron identity modeled only as a random intercept. This assumes every neuron has the same coupling slope. We tested this assumption by fitting random-slope mixed models that allow the $L/\lambda \rightarrow$ spatial metric slope to vary across neurons (Methods).

The coupling slope varies substantially across neurons. For all three metrics, the random-slope model was overwhelmingly preferred over the random-intercept-only model (likelihood ratio test: $\chi^2 = 1{,}724$, $p < 10^{-300}$ for Clark-Evans; $\chi^2 = 23{,}629$, $p < 10^{-300}$ for Interval CV; $\chi^2 = 1{,}021$, $p = 1.6 \times 10^{-222}$ for compactness; all with 2 df). The regime coupling is not a fixed property of the cortex — it varies per neuron.

Interval CV reveals a qualitative cell-type dissociation. The most striking result emerged from the Interval CV slopes. We extracted per-neuron BLUP estimates and compared their distributions between excitatory and inhibitory neurons (Table 3).

Excitatory neurons show steep negative slopes: as L/λ increases, inter-synapse spacing becomes dramatically more uniform (median slope $= -2.23$, $n = 1{,}702$ neurons). This means that for excitatory dendrites, electrotonic isolation is a strong signal for how to arrange synapses — isolated branches receive more regularly spaced inputs.

Inhibitory neurons, by contrast, show near-zero slopes (median slope $= -0.07$, $n = 436$ neurons). Inter-synapse spacing regularity barely changes with electrotonic regime. This is not a power issue — the difference between cell types is massive (Mann-Whitney $U = 253{,}095$, $p = 1.1 \times 10^{-24}$, Cohen's $d = 0.83$).

This is not simply "inhibitory coupling is stronger than excitatory." It is a qualitatively different organizational logic. Excitatory neurons *use* electrotonic regime as a signal for synaptic spacing regularity: the more isolated the branch, the more uniform the intervals. Inhibitory neurons maintain relatively uniform inter-synapse spacing *regardless* of how electrotonically isolated the branch is. The coupling detected in the fixed-effects models (Section 2.4) is real but operates through a different channel — primarily through nearest-neighbor regularity (Clark-Evans) and global dispersion (compactness), not through interval uniformity.

Clark-Evans and compactness slopes. Clark-Evans slopes showed a modest but significant cell-type difference (excitatory median $= 0.91$, inhibitory median $= 1.14$; $p = 1.2 \times 10^{-4}$, $d = 0.16$). Compactness slopes did not differ between cell types (excitatory median $= 2.89$, inhibitory median $= 2.98$; $p = 0.21$, $d = -0.28$). The fact that global synapse dispersion (compactness) follows the same L/λ slope in both cell types, while local spacing regularity (Interval CV) does not, reinforces the distinction between the two spatial dimensions identified in the partner mediation analysis (Section 2.5).

Random slope variance. The random slope variance was largest for Interval CV ($\sigma^2_{\text{slope}} = 22.2$) — 17× larger than Clark-Evans ($\sigma^2_{\text{slope}} = 1.28$) and 4× larger than compactness ($\sigma^2_{\text{slope}} = 5.22$). This asymmetry reflects the qualitative cell-type split: because excitatory and inhibitory neurons have dramatically different Interval CV slopes (median −2.23 vs. −0.07), the population-level variance in this slope is inflated by the bimodal distribution across cell classes. Clark-Evans and compactness slopes, which differ less between cell types, show correspondingly less between-neuron variance. The Interval CV slope is therefore not simply the "noisiest" metric — it is the metric that most sharply discriminates the two organizational logics.

Reconciling fixed-effects and random-slope results. A careful reader may note an apparent tension: the fixed-effects model (Section 2.4) shows inhibitory regime coupling ∼2× stronger than excitatory, yet the random-slope analysis shows inhibitory Interval CV slopes near zero. These are not contradictory. The strong inhibitory coupling in the fixed-effects model is driven primarily by Clark-Evans (nearest-neighbor regularity) and compactness (global dispersion), where inhibitory neurons do show steeper slopes. The Interval CV channel — inter-synapse interval uniformity — is the specific dimension on which inhibitory neurons decouple from electrotonic regime. The fixed-effects model, which estimates an aggregate regime coefficient, averages across these channels and obscures the metric-specific dissociation that the random-slope analysis reveals.

2.7 Pairwise compactness non-monotonicity is a normalization artifact

While Clark-Evans and Interval CV show clean monotonic dose-response patterns, pairwise compactness shows a non-monotonic relationship with electrotonic regime: regime 1 (intermediate) branches show *less* compact synapses (mean $z = -0.27$) than either regime 0 ($z = -0.65$) or regime 2 ($z = -0.88$). This non-monotonicity is consistent across all cell types, ruling out a subtype-specific artifact (Figure 6).

We investigated three candidate explanations: branch length confounds, synapse density differences, and regime × branch length interactions.

Branch length differs dramatically by regime. By construction, branches in different regimes have different electrotonic — and therefore physical — lengths. Regime 0 branches have a median length of 11.6 μm, regime 1 of 42.3 μm, and regime 2 of 104.7 μm (Kruskal-Wallis $H = 110{,}679$, $p < 10^{-300}$). Synapse density (synapses per nm) also differs modestly across regimes ($H = 555$, $p = 3.0 \times 10^{-121}$), but not in a pattern that explains the non-monotonicity.

Compactness residuals reveal a regime × length interaction. After regressing out synapse count and total branch length from pairwise compactness z, the residuals still differ by regime: regime 0 mean residual = −0.37, regime 1 = +0.21, regime 2 = +0.11 (Kruskal-Wallis $H = 3{,}926$, $p < 10^{-300}$). Crucially, a regime × branch length interaction model revealed significant interactions for both regime 1 ($\beta = 5.2 \times 10^{-5}$, $p < 10^{-300}$) and regime 2 ($\beta = 4.8 \times 10^{-5}$, $p < 10^{-300}$). This means that the relationship between compactness and branch length changes slope at different regimes — the normalization inherent in z-scoring against a length-matched null does not fully remove the length dependence when the metric's sensitivity to length is regime-specific.

Interpretation. The non-monotonicity is a measurement artifact, not a biological transition zone. Pairwise compactness normalizes by the expected spread of uniformly distributed synapses on a branch, which itself scales with branch length. When the relationship between raw compactness

and branch length has different slopes in different regimes — as it does, because intermediate-length branches have specific geometric properties — the z-scored metric can show non-monotonic behavior even if the underlying spatial organization is monotonic. We therefore present Clark-Evans and Interval CV as the primary evidence for graded regime coupling, with pairwise compactness treated as a secondary metric whose population-level non-monotonicity reflects metric sensitivity to branch geometry rather than a biological phenomenon.

2.8 Partner-level validation of regime coupling

Two additional analyses confirm that the regime coupling reflects genuine partner-level wiring differences rather than a compositional artifact of input type.

Input balance does not explain regime coupling. The excitatory fraction of presynaptic inputs varies systematically across the dendritic tree, with distal (high-L/λ) branches typically receiving a different ratio of excitatory to inhibitory inputs than proximal branches. If this compositional gradient drives the spatial organization differences between regimes, controlling for excitatory fraction should eliminate the regime effect. Including excitatory fraction as a covariate reduced the regime 2 coefficient by 1.4% for Interval CV, 6.5% for Clark-Evans, and increased it by 7.0% for pairwise compactness (Table 4). The regime effect remained overwhelmingly significant in all cases ($p < 10^{-300}$). For inhibitory neurons, the mediation percentages were negligible to negative (-0.2% to -6.8%), indicating that controlling for input balance slightly *increases* the apparent regime effect. The coupling is therefore not a secondary consequence of input type composition.

Partner spatial spread scales with regime. To assess how individual presynaptic partners distribute their synapses across branches of different electrotonic regimes, we analyzed 178,103 partners making ≥ 3 synapses each. Partners innervating compartmentalized (regime 2) branches showed a mean pairwise spread of 91.3 μm, compared to 42.1 μm for partners on compact (regime 0) branches — a 2.17-fold difference (Figure 5). This confirms that the branch-level spatial organization patterns are accompanied by partner-level architectural differences: the same presynaptic neuron distributes its synapses more widely when contacting isolated dendritic domains.

2.9 Afferent subtype composition varies systematically with electrotonic regime

The inhibitory residual identified in Section 2.5 — the $\sim$86% of inhibitory coupling not attributable to partner architecture — could arise from cell-type-specific afferent targeting: if different presynaptic classes preferentially innervate branches of different electrotonic regimes, this compositional gradient could organize synaptic spacing through mechanisms orthogonal to generic partner diversity and contact multiplicity. We tested this by computing, for each dendritic branch of each inhibitory neuron, the fraction of typed presynaptic synapses originating from each inhibitory subtype (BC, MC, BPC, NGC) and from excitatory neurons (Methods).

BC afferents are strongly enriched on compact branches. Using a depth-preserving within-neuron shuffle null (1,000 permutations per neuron, preserving branch geometry and total per-class synapse counts within soma-distance tertiles; Methods), we found that basket cell afferents are strongly enriched on regime 0 (compact) branches: $\log_2$ enrichment $= 0.91$, $p < 0.001$ (permutation). At the neuron level, 88.9% of 434 inhibitory neurons showed higher BC fraction on compact than non-compact branches (median $\log_2$ ratio $= 0.99$, Hodges–Lehmann estimator $= 0.98$, Wilcoxon signed-rank $p = 4.0 \times 10^{-45}$, 95% bootstrap CI $= [0.92, 1.07]$). The effect was consistent

across all four postsynaptic subtypes: BC postsynaptic neurons (97.7% positive, $n = 213$), MC (87.7%, $n = 114$), BPC (73.5%, $n = 83$), and NGC (70.8%, $n = 24$).

Within-neuron targeting tests confirmed that this is not a between-neuron confound: 190 of 213 individual BC neurons showed significant within-neuron BC afferent targeting ($p < 0.05$, chi-squared test), with 208/213 showing the expected directional pattern (BC→low regime, MC→high regime).

MC targeting is real but modest. Martinotti cell afferents showed a statistically reliable but substantially weaker enrichment on compartmentalized branches (neuron-level median $\log_2$ ratio = 0.26, Hodges–Lehmann = 0.30, Wilcoxon $p = 2.3 \times 10^{-19}$, 67.4% positive). The BC effect is approximately 4× larger than the MC effect, consistent with the stronger compartment specificity of perisomatic-targeting interneurons compared to distal-targeting Martinotti cells (Markram et al., 2004; Schneider-Mizell et al., 2023).

BC targeting concentrates in proximal dendritic compartments. Stratifying by soma distance (per-neuron median split), the BC targeting effect was 16× stronger in proximal compartments (mixed-model regime 2 coefficient = -0.30, $p = 6.2 \times 10^{-147}$) than in distal compartments (coefficient = -0.019, $p = 0.036$). This proximal concentration is consistent with the perisomatic targeting biology of basket cells (Schneider-Mizell et al., 2023) but resolves finer electrotonic structure than compartment-level analyses alone: within proximal dendrites, BC afferents still preferentially target electrotonically compact branches.

Mixed-effects models confirmed the targeting pattern at population level: frac_BC_afferent $\sim C$(regime), with regime 2 coefficient = -0.28 ($p < 10^{-300}$, $n = 25{,}902$ branches, 436 neurons), and frac_MC_afferent: regime 2 coefficient = $+0.073$ ($p = 4.1 \times 10^{-39}$).

2.10 Afferent identity and partner architecture jointly explain 18–26% of inhibitory coupling

To determine whether afferent subtype composition adds explanatory power beyond the partner architecture features identified in Section 2.5, we fit a four-model comparison for each spatial metric (Table 5):

- **M0** (baseline): $z_{\text{metric}} \sim C(\text{regime}) + \text{synapse_count} + \text{total_length} + (1|\text{neuron})$
- **M1** (+ composition): M0 + frac_inh + frac_BC + frac_MC
- **M2** (+ partner architecture): M0 + unique_partners + mean_syn/partner + Gini + frac_multisynaptic
- **M3** (+ both): M0 + all composition and partner architecture covariates

For all three spatial metrics, M3 > M1 and M3 > M2: the combined model consistently explained more variance than either pathway alone. Combined mediation (M3) reached 23.3% for Clark-Evans z, 18.1% for Interval CV z, and 25.5% for pairwise compactness z — up from 14.2%, 13.3%, and 15.4% for partner architecture alone (M2). All bootstrap 95% CIs excluded zero (200 cluster-bootstrap iterations by neuron; Table 5).

Two partially independent pathways. The sequential decomposition reveals two pathways through which afferent identity organizes synaptic spacing:

The overlap pathway (4–6%): afferent composition acts *through* partner architecture. BC targeting determines which presynaptic neurons arrive, which determines contact multiplicity and partner diversity, which determines spacing. This was confirmed directly: BC fraction significantly predicted all four partner architecture features in mixed models (Link 1 mediation: 13–39%; BC fraction coefficients $p < 10^{-18}$ for all four features). BC-rich branches have fewer unique partners ($\beta = -2.20$, $p = 6.9 \times 10^{-19}$), higher contact multiplicity ($\beta = +0.064$, $p = 6.4 \times 10^{-83}$), higher partner concentration ($\beta = +0.039$, $p = 2.0 \times 10^{-83}$), and more multisynaptic contacts ($\beta = +0.043$, $p = 1.5 \times 10^{-85}$) — precisely the partner architecture profile that drives compact spacing.

The chain incremental pathway (5–10%): afferent composition explains regime variance in spatial metrics *beyond* what partner architecture captures. This placement effect indicates that different presynaptic classes impose different spacing rules conditional on where they synapse, a hypothesis tested directly in Section 2.11.

Subtype-specific chain patterns. Stratifying by postsynaptic subtype confirmed that the chain incremental is concentrated in BC (5–11%) and MC (5–9%) postsynaptic neurons, while BPC neurons are dominated by partner architecture alone (M2 = 21–67%, chain incremental = 1–5%). This dissociation indicates that the mechanisms differ by postsynaptic subtype: BC and MC postsynaptic neurons benefit from afferent-specific placement rules, while BPC organization is largely explained by generic partner architecture.

2.11 Presynaptic subtype imposes distinct within-axon placement rules

The chain incremental (Section 2.10) demonstrates that afferent identity affects spatial organization beyond partner architecture, but does not reveal whether this reflects a genuine placement effect or a residual confound. We tested this directly with an axon-conditioned placement analysis: for multisynaptic typed partners (same presynaptic neuron with ≥ 2 synapses on the same branch), we computed the mean pairwise 3D distance between their synapses and tested whether electrotonic regime predicts this within-partner placement metric after conditioning on contact multiplicity and branch length (Methods).

BC contacts are placed more compactly on compact branches. Among 7,478 BC partner-branch records (407 neurons), the mean within-partner pairwise distance increased monotonically with electrotonic regime: 2.5 μm (regime 0, n = 2,560), 3.3 μm (regime 1, n = 2,069), and 4.4 μm (regime 2, n = 2,849) — a 1.76-fold expansion from compact to compartmentalized. In a mixed-effects model controlling for contact count (k) and branch length, the regime 2 coefficient was +0.95 ($p = 1.1 \times 10^{-107}$; Table 6).

The effect survives exact multiplicity matching. To eliminate any residual multiplicity confound, we analyzed partners at each contact count separately. The regime gradient persisted at every k value tested: $k = 2$ (regime 2 coefficient = 0.93, $p = 1.8 \times 10^{-67}$, n = 5,578), $k = 3$ (= 1.07, $p = 4.7 \times 10^{-39}$, n = 1,274), and $k = 4$ (= 1.00, $p = 4.1 \times 10^{-18}$, n = 407). This eliminates the possibility that the placement effect is an artifact of differential contact multiplicity across regimes.

Neuron-level confirmation. At the neuron level, 81.0% of 300 neurons with BC partners in both regime 0 and regime 2 showed wider BC contact placement on compartmentalized branches

(median $\log_2(\mathrm{R2/R0}) = 0.90$, Hodges–Lehmann = 0.97, Wilcoxon $p = 1.2 \times 10^{-31}$, 95% CI = $[0.76, 1.08]$).

MC contacts spread more broadly than BC contacts on compartmentalized branches. MC partners showed a qualitatively similar but quantitatively larger expansion: median within-partner distance increased from 2.7 μm (regime 0) to 8.1 μm (regime 2) — a 3.0-fold expansion compared to BC's 1.76-fold. An interaction model (regime $\times$ presynaptic class, controlling for k and branch length) confirmed that the regime 2 $\times$ MC interaction was significant ($\beta = +0.24$, $p = 5.2 \times 10^{-4}$), meaning MC contacts spread significantly more on compartmentalized branches than BC contacts do, conditional on being there.

This subtype-specific placement difference — BC tight, MC broad — aligns with the known axonal morphology of these classes: basket cell axons form dense perisomatic bouton clusters, while Martinotti cell axons arborize more broadly in distal dendritic domains (Markram et al., 2004). The axon-conditioned analysis shows that this morphological difference manifests at the level of individual branch electrotonic structure, not just at the coarse compartment level.

Mechanistic interpretation. Together, Sections 2.9–2.11 establish a complete mechanistic chain:

electrotonic regime $\rightarrow$ *afferent subtype composition* $\rightarrow$ *partner architecture* $\rightarrow$ *within-axon placement* $\rightarrow$ *synaptic spacing*

Each link is independently validated. The arrival effect (Section 2.9) determines which presynaptic classes innervate a branch. The partner architecture effect (Section 2.5) determines the diversity and multiplicity of contacts. The placement effect (this section) determines where individual axons position their boutons. These three mechanisms are partially independent: their combined mediation (M3 = 18–26%) exceeds either pathway alone, and the placement effect survives exact multiplicity conditioning.

3 Discussion

3.1 Two wiring mechanisms, not one

The partner architecture mediation analysis resolves the mechanistic shallowness of the initial population-scale replication. Where input balance (excitatory fraction) mediated only 1–10% of the regime coupling, partner architecture features mediate 25–53%, establishing partner-level innervation structure as the dominant mechanistic layer.

The dissociation between which features mediate which spatial dimensions is not redundancy — it identifies two distinct wiring mechanisms. *Partner diversity* (the number of unique presynaptic neurons contacting a branch) primarily governs local spacing regularity: branches innervated by more distinct partners show more evenly spaced synapses. This is consistent with a geometric packing constraint — more partners competing for postsynaptic territory produce more regular spacing through mutual exclusion (Kasthuri et al., 2015). *Contact multiplicity* (how many synapses each partner makes per branch) primarily governs global synapse dispersion: branches where individual partners make repeated contacts show more concentrated synapse populations. This is consistent with the "multi-contact" wiring rule, in which axons that have already established one contact tend to make additional contacts nearby (Druckmann et al., 2014).

These two mechanisms can coexist because they operate at different spatial scales: partner diversity sets the macro-scale regularity of the synapse population, while contact multiplicity determines the micro-scale clustering of individual partner contributions. The mediation hierarchy — partner diversity for Clark-Evans and Interval CV, contact multiplicity for compactness — maps directly onto the local-spacing versus global-dispersion distinction identified in the metric design (Section 2.1).

3.2 Decomposing the inhibitory residual

The most consequential finding from the mediation analysis is the asymmetry between cell types. Partner architecture mediates 29–53% of excitatory coupling but only 12–14% of inhibitory coupling, despite inhibitory neurons showing $\sim$2$\times$ stronger regime effects. The afferent decomposition (Sections 2.9–2.11) partially resolves this asymmetry, reducing the inhibitory residual from $\sim$86% to $\sim$75% by identifying a multi-stage mechanistic chain.

The chain has three independently validated links. First, electrotonic regime predicts afferent subtype composition: basket cell afferents concentrate on compact branches with a neuron-level effect that is consistent across 89% of inhibitory neurons. This targeting is not merely a depth effect — it persists within proximal dendritic compartments, resolving finer electrotonic structure than compartment-level analyses (Schneider-Mizell et al., 2023). Second, afferent subtype composition predicts partner architecture: BC-rich branches show the precise partner profile — fewer unique partners, higher contact multiplicity, greater partner concentration — that drives compact spacing. Third, presynaptic subtype identity imposes distinct within-axon placement rules that survive exact multiplicity conditioning. This third link is the most novel: it demonstrates that the spatial organization of inhibitory synapses reflects not only *who* arrives and *how often*, but also *how* each presynaptic class positions its boutons once it has arrived.

The $\sim$75% that remains unexplained likely reflects cell-autonomous mechanisms not accessible through connectomic data: contact-mediated synapse repulsion, developmental timing constraints, or molecular signals that enforce regular spacing independently of presynaptic identity (Spruston, 2008). The random-slope analysis (Section 2.6) supports this interpretation: inhibitory neurons maintain near-uniform inter-synapse spacing regardless of electrotonic regime (median slope $= -0.07$), consistent with a spacing mechanism that operates independently of the branch's biophysical properties.

3.3 Distinct organizational logics for excitatory and inhibitory spacing

The random-slope Interval CV result transforms the cell-type dissociation from a quantitative difference ("inhibitory coupling is 2$\times$ stronger") into a qualitative one. Excitatory neurons use electrotonic regime as a graded signal for synaptic spacing regularity: the more isolated the branch, the more uniform the inter-synapse intervals (median slope $= -2.23$). Inhibitory neurons do not: their inter-synapse spacing is nearly independent of electrotonic regime (median slope $= -0.07$).

This distinction has implications for models of synaptic organization. For excitatory neurons, the strong $L/\lambda \rightarrow$ spacing regularity coupling is consistent with a model in which electrotonic isolation is associated with specific spacing patterns. One possibility is that on isolated branches, where nonlinear integration depends on the precise timing and location of inputs (Branco et al., 2010; Larkum et al., 2009), uniform spacing distributes inputs across the branch's integrative zone, while on compact branches, where inputs sum linearly regardless of location, uniform spacing confers no advantage. However, this functional interpretation goes beyond what our structural data can establish; the coupling could equally reflect accessibility-driven wiring constraints that

correlate with electrotonic isolation.

For inhibitory neurons, the near-independence of spacing regularity from electrotonic regime suggests a different organizational principle. One possibility is that inhibitory synapse placement is governed by a cell-autonomous spacing mechanism — for example, contact-mediated repulsion between synapses (Spruston, 2008) — that produces regular spacing on all branches regardless of their biophysical properties. Under this model, the strong regime coupling detected by the fixed-effects model in inhibitory neurons operates entirely through the nearest-neighbor regularity (Clark-Evans) and global dispersion (compactness) channels, not through interval uniformity.

The fact that compactness slopes do not differ between cell types ($p = 0.21$) reinforces this interpretation. Global synapse dispersion responds identically to electrotonic regime in both cell types, suggesting that it is governed by universal physical constraints (e.g., axonal reach, branch length). The cell-type specificity is confined to local spacing regularity — the dimension most likely to reflect active biological regulation rather than passive geometry.

3.4 Resolving the compactness non-monotonicity

The non-monotonic dose-response in pairwise compactness, previously a loose end suggesting a possible biological "transition zone" at intermediate electrotonic lengths, is resolved as a normalization artifact. The regime $\times$ branch length interaction demonstrates that the compactness z-score's sensitivity to branch length differs across regimes, producing apparent non-monotonicity in the z-scored metric even when the underlying spatial organization varies monotonically with L/λ.

This does not invalidate pairwise compactness as a metric — it remains informative about global synapse dispersion within a regime. But it means that cross-regime comparisons of compactness z-scores should be interpreted cautiously when the regimes differ substantially in branch length, as they do by construction. Clark-Evans and Interval CV are less susceptible to this artifact because nearest-neighbor and interval metrics are less sensitive to branch geometry than pairwise distance metrics.

3.5 Relation to the original 12-neuron study

The present results both confirm and extend the original finding of 69% partner-mediated coupling in excitatory neurons (Voruganti, 2026). At population scale, partner architecture mediates 29–53% of excitatory coupling, depending on the metric. The lower bound is expected: the original study used a single feature (mean synapses per partner) with a more aggressive matching procedure on 12 manually curated neurons, while the present study uses four features across 1,702 excitatory neurons with automated matching. The convergence on the same order of magnitude — tens of percent, not single digits — validates the original mechanism while providing more precise and generalizable estimates.

The inhibitory mediation finding (12–14%) is novel. The original study estimated 39% mediation for inhibitory neurons, but with $n = 2$, this estimate had wide uncertainty. The population-scale result establishes that inhibitory coupling is genuinely less mediated by partner architecture than excitatory coupling, confirming the cell-type asymmetry as a robust biological property rather than a small-sample artifact.

3.6 Limitations

Several limitations should be noted. First, all neurons are from a single MICrONS cortical volume (mouse visual cortex area VISp). While the within-volume replication is comprehensive (all 2,138

proofread neurons), cross-dataset generalization — to other cortical areas, other species (e.g., H01 human cortex), or subcortical structures — remains untested.

Second, electrotonic lengths are computed from EM-measured diameters, which are subject to tissue shrinkage during electron microscopy preparation. The absolute L/λ values are therefore approximate. However, the sensitivity analysis demonstrates that the coupling is invariant to threshold choice across a wide range, and the continuous analysis uses relative L/λ rankings rather than absolute values.

Third, the passive cable parameters (R_M, R_I) are fixed at standard values. In vivo, active conductances, synaptic conductance loads, and neuromodulation alter the effective electrotonic length of a branch dynamically. Our analysis captures the static structural regime, not the instantaneous functional regime.

Fourth, the afferent subtype labels (BC, MC, BPC, NGC) are morphological EM proxy categories, not molecular identity classes. The correspondence with molecular types (PV, SST, VIP, 5HT3aR) is approximate (Schneider-Mizell et al., 2023): our claims about presynaptic identity should be interpreted as "consistent with" known molecular targeting biology rather than direct demonstrations of molecular identity.

Fifth, of all presynaptic synapses, only ∼18% have resolved cell type annotations; among multi-synapse partners ($k \geq 3$), coverage reaches ∼43%. The afferent composition analyses therefore operate on the typed subset, which may not be fully representative of all inputs. Coverage is regime-asymmetric (proximal branches have lower subtyped coverage than distal), but this asymmetry works *against* finding BC enrichment on compact branches — making the observed effect conservative.

Sixth, the combined mechanistic chain (afferent identity + partner architecture) explains 18–26% of inhibitory coupling, leaving ∼75% unexplained. Candidate mediators not tested include cell-autonomous spacing mechanisms, developmental timing, axonal geometry constraints, and molecular signals inaccessible to connectomic data.

Fifth, the random-slope model AIC/BIC values were numerically undefined (NaN) due to the large sample size and model complexity, though the likelihood ratio tests and parameter estimates were valid. The model comparison should be interpreted through the LRT statistics rather than information criteria.

Finally, we do not link spatial organization to neural computation. Whether the spatial patterns have functional consequences — for example, whether the excitatory spacing-regularity coupling produces different patterns of dendritic integration than the inhibitory cell-autonomous spacing — requires combined structural and functional data (Jia et al., 2010).

4 Methods

4.1 Data acquisition

We queried the MICrONS mouse visual cortex EM volume (MICrONS Consortium, 2021; Turner et al., 2022) using CAVEclient (v8.0.1) with the `minnie65_public` datastack. Cell type annotations were obtained from the AIBS metamodel cell types table (v661); proofreading status from the proofreading status table. All 2,178 proofread neurons were downloaded; 2,138 passed quality control (see below). Dendritic skeletons with per-node radii were obtained via the skeleton service. Postsynaptic synapse positions were queried from the `synapses_pni_2` table. Presynaptic partner cell types were resolved by batch-querying the cell type table for unique presynaptic root IDs. Voxel coordinates were converted to nanometers using the volume resolution $[4, 4, 40]$ nm/voxel.

4.2 Quality control

Neurons were excluded if: (a) the skeleton failed to parse or had zero dendrite branches, (b) fewer than 20 synapses were recovered, (c) fewer than 80% of synapses snapped successfully to the skeleton (snap distance threshold 50 μm), or (d) fewer than 50% of presynaptic partners had resolved cell type annotations. Of 2,178 downloaded neurons, 2,138 passed all criteria (98.2%).

4.3 Branch morphometry and electrotonic length

Dendritic skeletons were parsed into branches at bifurcation points. For each branch, we computed physical length (sum of inter-node Euclidean distances in nm) and mean diameter (twice the mean of per-node radii from the EM reconstruction, in nm). Electrotonic length was computed as:

$$L/\lambda = \frac{L}{\sqrt{R_M \cdot d/(4R_I)}}$$

where L is physical length (cm), d is diameter (cm), $R_M = 20{,}000\ \Omega \cdot \mathrm{cm}^2$, and $R_I = 150\ \Omega \cdot \mathrm{cm}$ (Rall, 1967; Koch, 1999).

4.4 Regime classification

Branches were classified into three electrotonic regimes using tertile thresholds on the population L/λ distribution: regime 0 ($L/\lambda < 0.037$, compact), regime 1 ($0.037 \leq L/\lambda < 0.157$, intermediate), and regime 2 ($L/\lambda \geq 0.157$, compartmentalized). For sensitivity analysis, we additionally tested biophysical thresholds ($L/\lambda = 0.1, 0.5$) and six quantile-based thresholds spanning 20th/80th to 40th/60th percentile splits.

4.5 Spatial organization metrics

For each branch with ≥ 3 synapses, we computed three spatial metrics. The first two capture local spacing structure (are neighboring synapses evenly distributed?), while the third captures global dispersion (how spread out is the entire synapse population?).

Clark-Evans ratio (Clark and Evans, 1954): the ratio of mean observed nearest-neighbor distance to the expected mean under a uniform distribution on the branch path. Values > 1 indicate regular spacing; < 1 indicates clustering.

Interval CV: the coefficient of variation of inter-synapse intervals along the branch path. Lower values indicate more uniform spacing.

Pairwise compactness: the mean Euclidean distance between all pairs of synapses on the branch. Lower values indicate tighter spatial concentration.

4.6 Null model and z-scoring

Each metric was z-scored against a dendrite-constrained null model. For each branch, we drew 1,000 independent samples of n uniformly distributed points along the branch path (where n is the observed synapse count) and computed the metric for each null draw. The z-score is $(x_{\text{obs}} - \mu_{\text{null}})/\sigma_{\text{null}}$.

4.7 Statistical testing

We fit linear mixed-effects models using the `statsmodels` MixedLM implementation in Python:

$$z_{\text{metric}} \sim \text{regime} + \text{synapse_count} + \text{total_length} + \text{exc_fraction} + (1|\text{neuron})$$

where regime is treated as a categorical variable (reference: regime 0), synapse count and total branch length are continuous covariates, excitatory fraction is the proportion of presynaptic partners classified as excitatory, and neuron identity is a random intercept. For continuous analyses, regime is replaced by continuous L/λ. Significance was assessed via Wald z-tests on fixed-effect coefficients.

Effect sizes were computed as Cohen's d between regime 0 and regime 2 z-score distributions. Dose-response monotonicity was assessed via Spearman rank correlation between regime label and metric z-score.

4.8 Partner architecture features

For each branch, presynaptic partner identities were assigned by matching synapse coordinates to presynaptic partner CSV files via KDTree nearest-neighbor lookup. For each branch with ≥ 3 snapped synapses, we computed: (1) **unique partners** — the count of distinct presynaptic root IDs; (2) **mean synapses per partner** — total synapse count divided by unique partners; (3) **Gini coefficient** — the Gini index of the per-partner synapse count distribution, measuring inequality (0 = all partners contribute equally, 1 = one partner dominates); (4) **fraction multi-synaptic** — the proportion of partners making ≥ 2 contacts on the branch. Features were computed for 127,697 branches across 2,138 neurons.

4.9 Mediation analysis

Mediation was tested using the coefficient-comparison approach (Baron and Kenny, 1986). For each mediator, we compared:

- **Reduced model:** $z_{\text{metric}} \sim \text{C(regime)} + \text{synapse_count} + \text{total_length} + (1|\text{neuron})$
- **Full model:** reduced model + mediator variable(s)

Mediation percentage was computed as $100 \times (1 - |\beta_{\text{full}}|/|\beta_{\text{reduced}}|)$, where β is the regime 2 coefficient. Positive values indicate that the mediator absorbs part of the regime effect; negative values indicate suppression (the mediator reveals additional regime effect). We note that mediation here is a statistical decomposition of covariance, not a demonstration of causal direction: partner architecture features could be causal mediators or correlated consequences of the same accessibility constraints that drive both regime assignment and spatial organization.

4.10 Random-slope mixed models

To test whether the $L/\lambda \rightarrow$ spatial coupling varies across neurons, we fit random-slope models:

$$z_{\text{metric}} \sim L/\lambda + \text{synapse_count} + \text{total_length} + (L/\lambda \,|\, \text{neuron})$$

This allows each neuron to have its own L/λ slope (BLUP estimate) while estimating a population-average fixed effect. The random-slope model was compared to the random-intercept-only model via likelihood ratio test (LRT; χ^2 with 2 df for the slope variance and intercept-slope covariance). Per-neuron slopes were extracted as the sum of the fixed effect and the neuron-specific random slope. Excitatory vs. inhibitory slope distributions were compared using Mann-Whitney U tests and Cohen's d.

4.11 Compactness geometry diagnostics

The non-monotonic pairwise compactness dose-response was investigated through: (1) comparison of branch length and synapse density distributions across regimes; (2) OLS residual analysis, regressing compactness z on synapse count and total branch length, then testing whether residuals still differ by regime; (3) interaction model, adding regime $\times$ branch length terms to test whether the compactness–length relationship has different slopes across regimes.

4.12 Partner spatial spread

For each presynaptic partner making ≥ 3 synapses onto a postsynaptic neuron, we computed the mean 3D Euclidean distance between all pairs of their synapse positions. Partners were classified by the dominant electrotonic regime of the branches they innervate (the regime containing the plurality of their synapses).

4.13 Afferent subtype composition

For each dendritic branch of each inhibitory neuron, we matched presynaptic synapses to partner identity using KDTree nearest-neighbor lookup against presynaptic partner CSV files. Each synapse was assigned an annotation level: L1 (exact subtype from CAVE annotation), L2 (subtype from catalog backfill), L3 (broad class only: excitatory/inhibitory), or L4 (unknown). Afferent fractions (frac_BC, frac_MC, frac_BPC, frac_NGC) were computed using L1+L2 synapses; broad fractions (frac_exc, frac_inh) used L1+L2+L3.

4.14 Depth-preserving shuffle null

For each postsynaptic neuron, we assigned each synapse to a soma-distance tertile bin (proximal/medial/distal, computed per neuron). Within each bin, presynaptic subtype labels were randomly permuted 1,000 times, preserving branch geometry, synapse count per branch, total per-class synapse counts, and depth structure. The observed afferent fraction per regime was compared to the null distribution. Sensitivity analysis with 2 and 4 depth bins confirmed that results were not bin-dependent.

4.15 Four-model mediation comparison

For each spatial metric, we fit four nested mixed-effects models on inhibitory branches: M0 (baseline: regime + synapse count + branch length + random intercept for neuron), M1 (M0 + afferent fractions), M2 (M0 + partner architecture features), and M3 (M0 + both). Mediation percentage was computed as $100 \times (1 - |\beta_{\text{full}}|/|\beta_{\text{reduced}}|)$. The chain incremental (= M3 − M2) measures the contribution of afferent composition beyond partner architecture; the overlap (= M1 + M2 − M3) measures the shared pathway. Bootstrap 95% CIs were computed via 200 cluster-bootstrap iterations (resampling neurons with replacement, fitting OLS).

4.16 Axon-conditioned placement analysis

For each inhibitory neuron, we identified multisynaptic typed partners: presynaptic neurons with ≥ 2 synapses of known subtype on the same branch. For each partner-branch pair, we computed the mean 3D pairwise Euclidean distance between all synapse pairs. The placement metric was modeled as: $\log(1+\text{mean_pairwise_dist}) \sim C(\text{regime}) + k_{\text{synapses}} + \text{total_length} + (1|\text{neuron})$. The

equal-k analysis stratified by exact contact count ($k = 2, 3, 4$) and refit within each stratum. The interaction model tested $C(\text{regime}) \times C(\text{pre_class})$ with BC as reference.

4.17 Software

All analyses used Python 3.9 with NumPy, pandas, SciPy, and statsmodels. Skeleton parsing and synapse snapping used custom code. Figures were generated with Matplotlib using the Wong (2011) colorblind-safe palette.

Table 1: Summary of neuron counts, branch counts, and Clark-Evans regime 2 coupling coefficients by cell type. Cohen's d computed between regime 0 and regime 2 Clark-Evans z-score distributions.

Class	Subtype	Neurons	Branches	Regime 2 coef.	Cohen's d	p
Inhibitory	BC	213	13,944	1.68	1.15	$< 10^{-290}$
	BPC	85	3,573	1.16	0.83	$< 10^{-37}$
	NGC	24	1,693	1.08	0.39	$< 10^{-12}$
	MC	114	7,438	0.79	0.64	$< 10^{-38}$
Excitatory	5P-ET	70	9,330	1.34	1.00	$< 10^{-126}$
	5P-NP	13	389	1.19	1.14	$< 10^{-5}$
	6P-CT	164	7,111	1.06	0.91	$< 10^{-80}$
	5P-IT	195	11,090	0.81	0.72	$< 10^{-64}$
	6P-IT	234	9,353	0.79	0.72	$< 10^{-51}$
	23P	518	48,328	0.68	0.50	$< 10^{-149}$
	4P	508	24,165	0.61	0.45	$< 10^{-79}$
All inhibitory		**436**	**26,648**	**1.32**	**0.94**	$< 10^{-300}$
All excitatory		**1,702**	**109,766**	**0.69**	**0.60**	$< 10^{-300}$

Table 2: Partner architecture mediation of regime coupling. Mediation %: proportion of the regime 2 coefficient absorbed by partner features. "Combined" includes all four features simultaneously. For comparison, excitatory fraction mediates only 1–10%.

Population	Metric	Individual best	Med. %	Combined med. %	n
Overall	Clark-Evans z	unique_partners	23.0%	28.9%	121,826
	Interval CV z	unique_partners	22.9%	25.2%	127,678
	Compactness z	mean_syn/partner	39.3%	43.3%	127,686
Excitatory	Clark-Evans z	frac_multisynaptic	23.7%	29.4%	97,000
	Interval CV z	unique_partners	29.0%	31.3%	102,387
	Compactness z	mean_syn/partner	48.5%	52.6%	102,394
Inhibitory	Clark-Evans z	unique_partners	13.2%	14.0%	24,826
	Interval CV z	unique_partners	11.2%	12.1%	25,291
	Compactness z	mean_syn/partner	15.0%	12.6%	25,292

Table 3: Random-slope mixed model results. Fixed slope: population-average L/λ coefficient. Slope variance: between-neuron variance in the L/λ slope. LRT compares random-slope to random-intercept-only model (2 df). Per-neuron slopes extracted as BLUPs.

Metric	Fixed slope	p	Slope var.	LRT χ^2	Exc. med.	Inh. med.	MW p
CE z	0.97	2.8×10^{-11}	1.28	1,724	0.91	1.14	1.2×10^{-4}
ICV z	−1.22	5.4×10^{-8}	22.24	23,629	−2.23	−0.07	1.1×10^{-24}
Comp. z	2.80	1.0×10^{-58}	5.22	1,021	2.89	2.98	0.21

Table 4: Mediation of regime coupling by excitatory/inhibitory input balance. Negative mediation values indicate that controlling for input balance increases the apparent regime effect.

Population	Metric	β_{reduced}	β_{full}	Mediation %
Overall	Clark-Evans z	0.794	0.743	6.5%
	Interval CV z	−1.720	−1.697	1.4%
	Compactness z	0.851	0.910	−7.0%
Excitatory	Clark-Evans z	0.765	0.690	9.8%
	Interval CV z	−1.359	−1.302	4.2%
	Compactness z	0.763	0.773	−1.2%
Inhibitory	Clark-Evans z	1.312	1.315	−0.2%
	Interval CV z	−2.673	−2.825	−5.7%
	Compactness z	1.694	1.810	−6.8%

Table 5: Four-model comparison for inhibitory regime coupling. M0: baseline (regime + controls). M1: + afferent composition. M2: + partner architecture. M3: + both. Mediation %: reduction in regime 2 coefficient relative to M0. Chain incremental = M3 − M2 (composition beyond partner architecture). Overlap = M1 + M2 − M3 (shared pathway). Bootstrap 95% CIs from 200 cluster-bootstrap iterations (in brackets).

Metric	M1 %	M2 %	M3 %	Chain	Overlap	n
Clark-Evans z	13.9	14.2	23.3 [17.7, 23.0]	9.1 [6.5, 10.1]	4.7 [2.8, 4.5]	24,826
Interval CV z	8.6	13.3	18.1 [13.2, 19.4]	4.9 [3.1, 6.2]	3.8 [2.4, 3.6]	25,291
Compactness z	16.6	15.4	25.5 [20.3, 25.9]	10.2 [8.8, 12.8]	6.4 [4.8, 6.8]	25,292

Table 6: Axon-conditioned placement analysis. Within-partner mean pairwise 3D distance for multisynaptic typed partners ($k \geq 2$ synapses on the same branch). Regime coefficients from mixed models controlling for k and branch length (log-transformed distance). Equal-k rows show the regime 2 coefficient when holding contact count exactly constant.

Class	Subset	n	Neurons	R0 (μm)	R2 (μm)	R2 coef	p
BC	All	7,478	407	2.5	4.4	+0.95	1.1×10^{-107}
	$k = 2$	5,578	403	2.1	3.7	+0.93	1.8×10^{-67}
	$k = 3$	1,274	304	3.0	5.7	+1.07	4.7×10^{-39}
	$k = 4$	407	191	3.6	7.4	+1.00	4.1×10^{-18}
MC	All	4,468	411	2.7	8.1	+0.79	8.3×10^{-20}
	$k = 2$	3,525	399	2.4	6.3	+0.77	1.2×10^{-13}
	$k = 3$	663	264	3.0	13.8	+0.88	6.8×10^{-7}
Exc	All	12,072	414	2.5	3.9	+0.46	1.7×10^{-21}

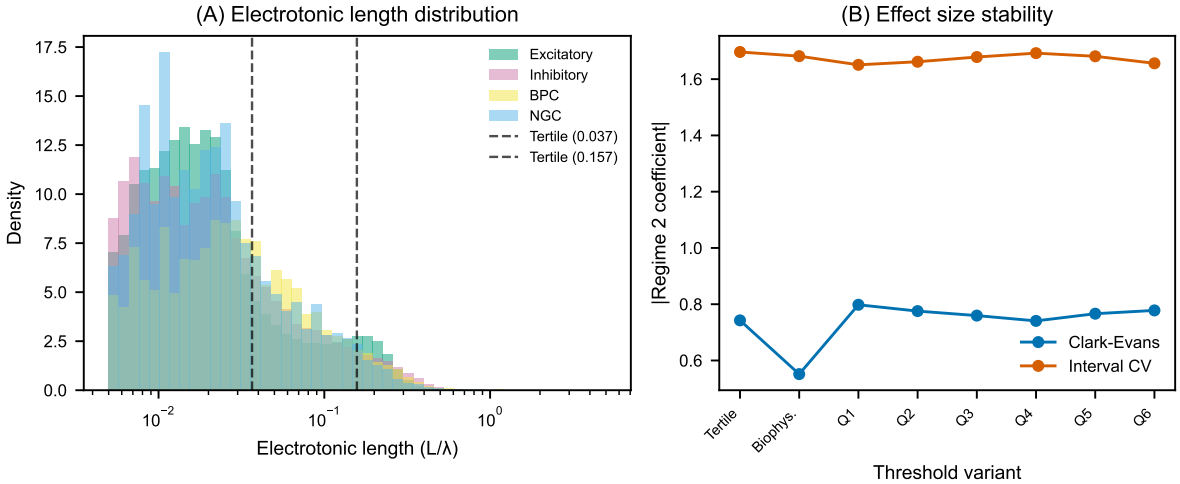

Figure 1: **Electrotonic length distribution and sensitivity analysis.** (A) Distribution of branch electrotonic lengths (L/λ) across 136,414 branches from 2,138 neurons. Vertical dashed lines indicate tertile thresholds. (B) Sensitivity of the regime 2 coefficient to threshold choice across eight discretization schemes for three spatial metrics. All 24 models converge, with Clark-Evans coefficients ranging from 0.55 to 0.80.

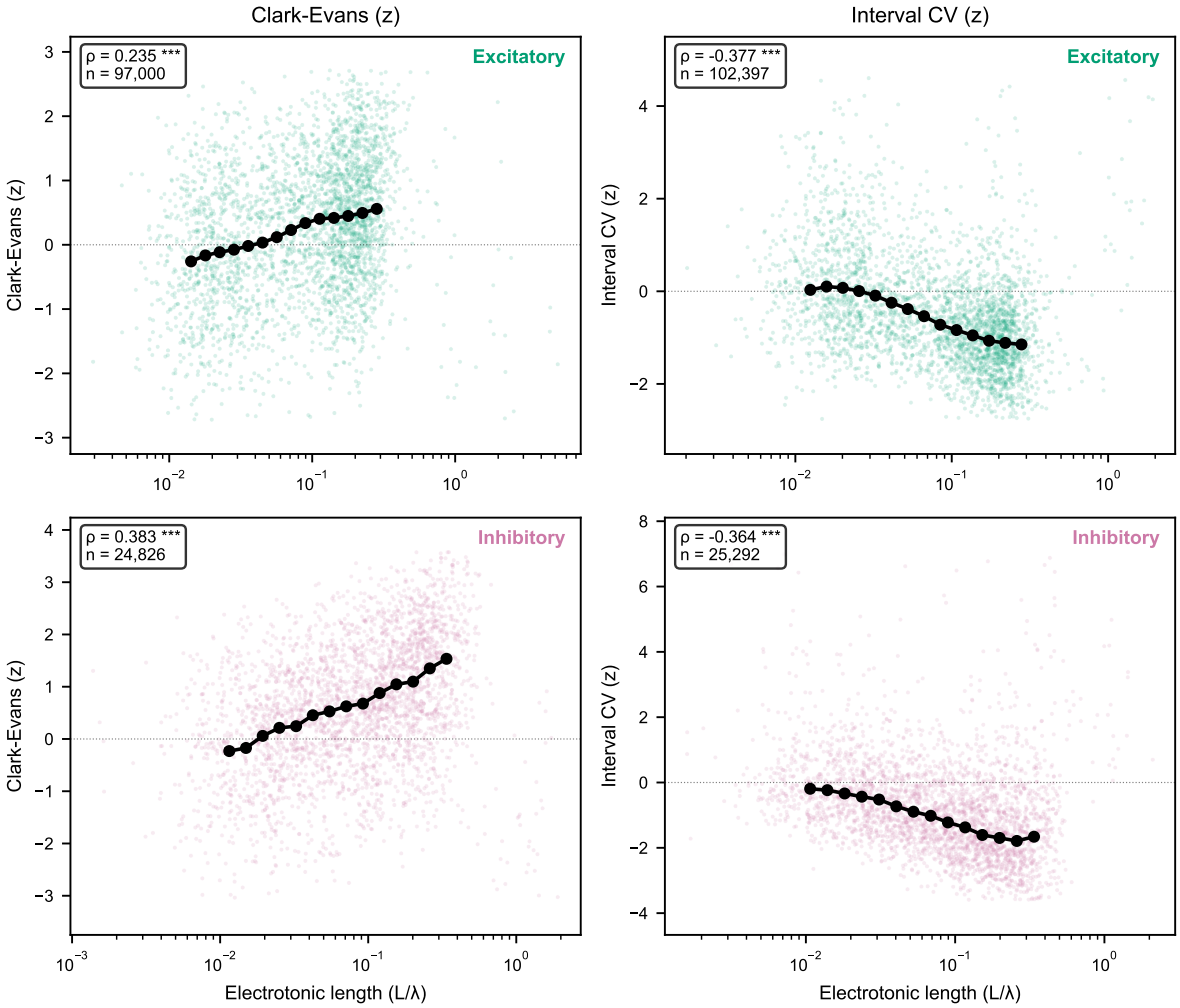

Figure 2: **Synaptic spatial organization tracks with continuous electrotonic length.** Scatter plots of L/λ vs. spatial metric z-scores for excitatory (top) and inhibitory (bottom) neurons, with binned medians (orange) computed in the 5th–95th percentile core (≥ 20 points per bin). Left: Clark-Evans z. Right: Interval CV z. Inhibitory neurons show steep, monotonic trends ($\beta = 7.65$, $p < 10^{-77}$ and $\beta = -17.5$, $p < 10^{-90}$, respectively). Excitatory neurons show significant but weaker continuous relationships.

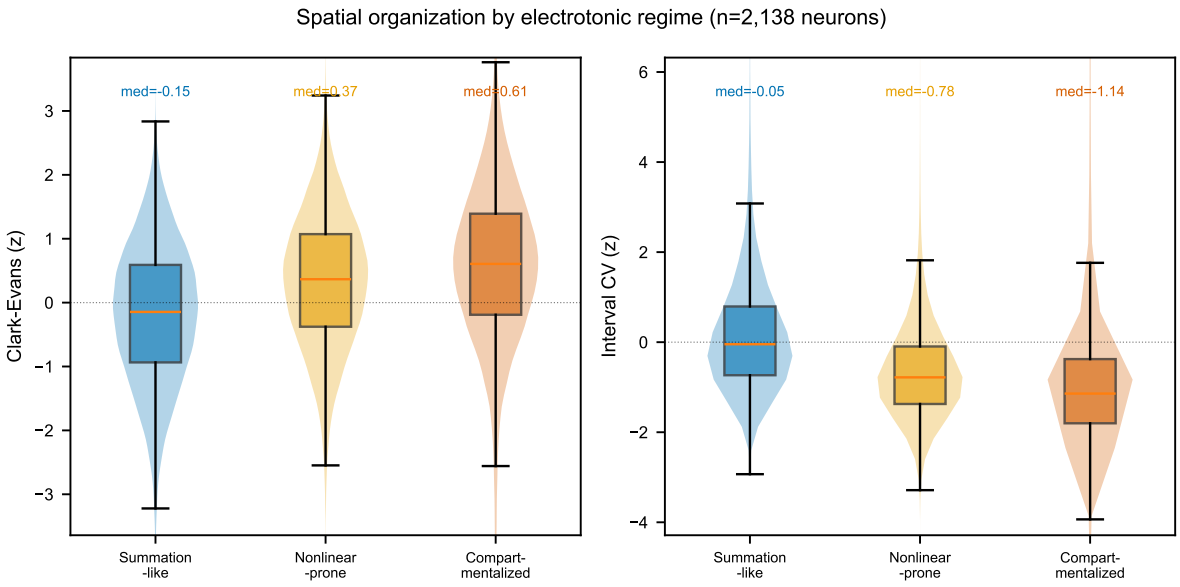

Figure 3: **Discretized regime analysis.** Violin plots of spatial metric z-scores stratified by electrotonic regime (0: compact, 1: intermediate, 2: compartmentalized) for 2,138 neurons. Clark-Evans z and Interval CV z show monotonic dose-response. Pairwise compactness z shows non-monotonic behavior explained by a regime $\times$ branch length normalization interaction (Section 2.7). All regime 2 vs. regime 0 comparisons: $p < 10^{-300}$.

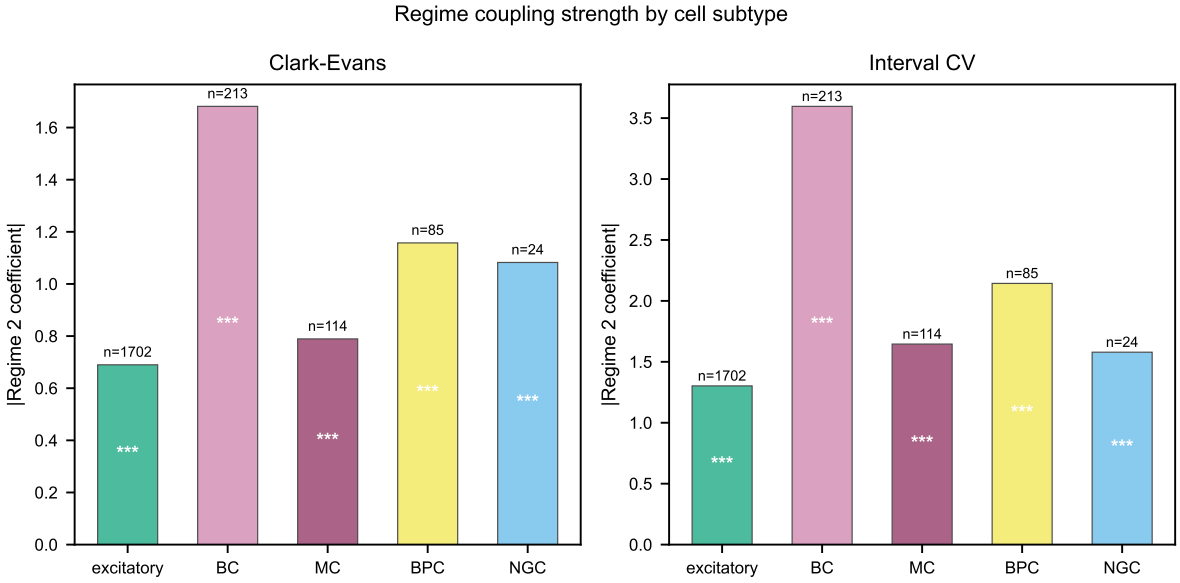

Figure 4: **Cell-type-specific coupling strength.** Clark-Evans regime 2 coefficients and Cohen's d effect sizes for 11 cell types. All four inhibitory subtypes (blue shades) show stronger coupling than the excitatory aggregate (orange). Basket cells are the strongest ($d = 1.15$); Martinotti cells are the weakest inhibitory subtype ($d = 0.64$) but still exceed the excitatory mean ($d = 0.60$). Error bars: 95% confidence intervals from mixed-effects models.

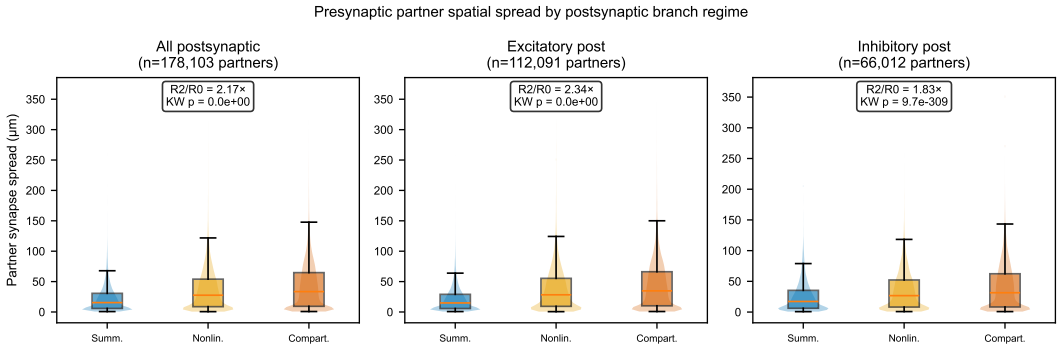

Figure 5: **Partner spatial spread by electrotonic regime.** Mean 3D pairwise distance of synapses from individual presynaptic partners ($k \geq 3$ synapses), classified by the dominant regime of innervated branches. Partners on regime 2 branches spread their synapses 2.17-fold wider than partners on regime 0 branches ($n = 178{,}103$ partners).

Supplementary Figures

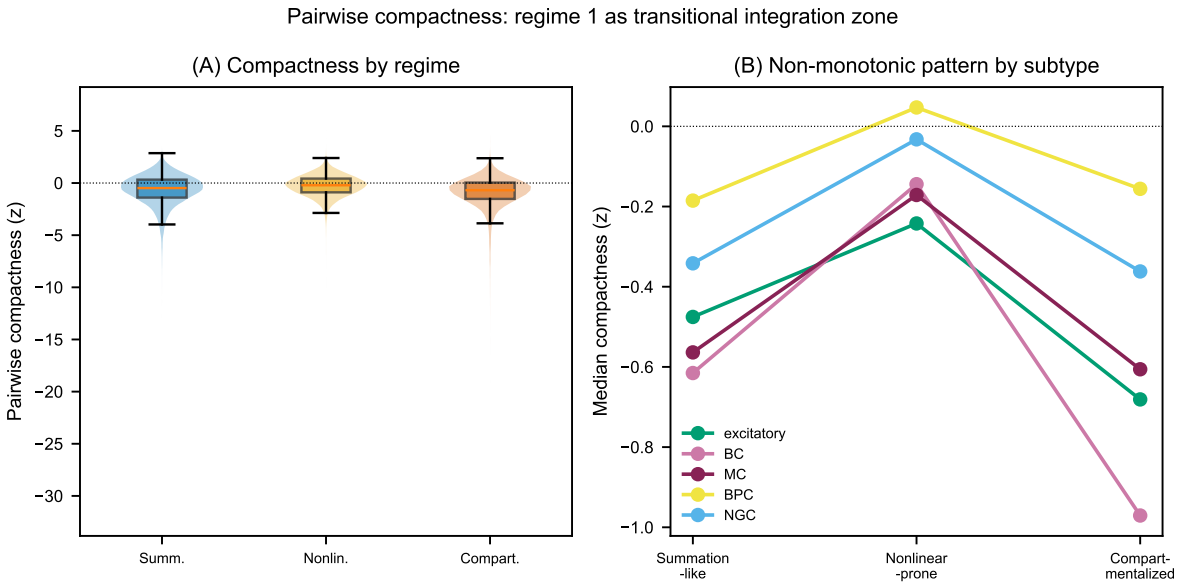

Figure 6: **Figure S1. Pairwise compactness non-monotonicity: normalization artifact across cell types.** Dose-response plots of median pairwise compactness z-score by electrotonic regime, stratified by cell type. The regime 1 reversal is consistent across excitatory and all four inhibitory subtypes, ruling out a subtype-specific artifact. Geometric diagnostics (Section 2.7) demonstrate that this non-monotonicity arises from a regime $\times$ branch length interaction in the z-score normalization: the compactness metric's sensitivity to branch length differs across regimes, producing apparent non-monotonicity in the normalized metric even when the underlying spatial organization varies monotonically with L/λ.

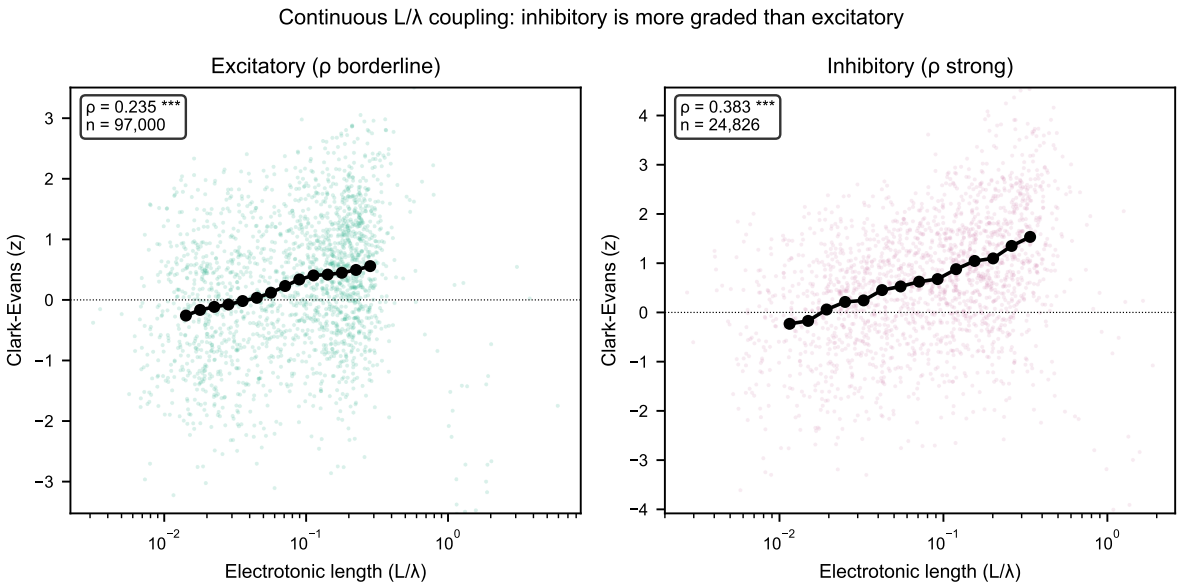

Figure 7: **Figure S2. Excitatory vs. inhibitory continuous coupling.** Direct comparison of L/λ vs. spatial metric z-scores between excitatory and inhibitory neurons. Inhibitory neurons show steeper slopes across all metrics. For pairwise compactness, the excitatory continuous relationship is null ($p = 0.67$) while the inhibitory relationship is strong ($p < 10^{-117}$), demonstrating that the metric-specific divergence is cell-type-dependent.

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